

# Chapter 3, Section 6

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1. Let  $(X, \mathcal{O}_X)$  be a ringed space, and let  $\mathcal{F}', \mathcal{F}'' \in \mathfrak{Mod}(X)$ . An *extension* of  $\mathcal{F}''$  by  $\mathcal{F}'$  is a short exact sequence

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

in  $\mathfrak{Mod}(X)$ . Two extensions are *isomorphic* if there is an isomorphism of the short exact sequences, inducing the identity maps on  $\mathcal{F}'$  and  $\mathcal{F}''$ . Given an extension as above consider the long exact sequence arising from  $\text{Hom}(\mathcal{F}'', \cdot)$ , in particular the map

$$\delta : \text{Hom}(\mathcal{F}'', \mathcal{F}'') \rightarrow \text{Ext}^1(\mathcal{F}'', \mathcal{F}'),$$

and let  $\xi \in \text{Ext}^1(\mathcal{F}'', \mathcal{F}')$  by  $\delta(1_{\mathcal{F}''})$ . Show that this process gives a one-to-one correspondence between isomorphism classes of extensions of  $\mathcal{F}''$  by  $\mathcal{F}'$ , and elements of the group  $\text{Ext}^1(\mathcal{F}'', \mathcal{F}')$ .

*Proof.* Let  $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{I}^\bullet$  be an injective resolution of  $\mathcal{F}'$ . An element of the group  $\xi \in \text{Ext}^1(\mathcal{F}'', \mathcal{F}')$  gives a morphism  $\xi : \mathcal{F}'' \rightarrow \mathcal{I}^1$  such that

$$\begin{array}{ccc} \mathcal{F}'' & & \\ \xi \downarrow & \searrow 0 & \\ \mathcal{I}^1 & \xrightarrow{\psi} & \mathcal{I}^2 \end{array}$$

commutes, i.e.,  $\xi \in \text{Hom}(\mathcal{F}'', \mathcal{I}^0/\mathcal{F}')$ . From now, denote this subsheaf of  $\mathcal{I}^1$  as  $\mathcal{J} = \mathcal{I}^0/\mathcal{F}'$ . The pull-back diagram of  $\xi$  and  $\psi$  gives the following commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}' & \longrightarrow & \mathcal{I}^0 \times_{\mathcal{I}^1} \mathcal{F}'' & \dashrightarrow & \mathcal{F}'' \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \xi \\ 0 & \longrightarrow & \mathcal{F}' & \longrightarrow & \mathcal{I}^0 & \xrightarrow{\psi} & \mathcal{J} \longrightarrow 0 \end{array},$$

so the top row is an extension of  $\mathcal{F}''$  by  $\mathcal{F}'$ . If we are given another representative  $\bar{\xi}$  of  $\xi$ , then  $\bar{\xi} - \xi = j \circ \psi$  for some  $\psi \in \text{Hom}(\mathcal{F}'', \mathcal{I}^0)$ , and repeating the construction above for  $j \circ \psi$  gives the trivial extension  $\mathcal{F}' \times \mathcal{F}''$  of  $\mathcal{F}''$  by  $\mathcal{F}'$ , so this construction is well-defined. Because  $\text{Hom}(\mathcal{F}'', \cdot)$  is a covariant  $\delta$ -functor, there is a commutative diagram

$$\begin{array}{ccc} \text{Hom}(\mathcal{F}'', \mathcal{F}'') & \xrightarrow{\delta} & \text{Ext}^1(\mathcal{F}'', \mathcal{F}') \\ \downarrow & & \downarrow \\ \text{Hom}(\mathcal{F}'', \mathcal{J}) & \longrightarrow & \text{Ext}^1(\mathcal{F}'', \mathcal{F}') \end{array}$$

which ensures  $\delta(1_{\mathcal{F}''}) = \xi$ . Hence, this process is surjective.

To prove injectivity, we show  $\mathcal{F} \cong \mathcal{I}^0 \times_{\mathcal{I}^1} \mathcal{F}''$ . Let  $\alpha : \mathcal{F} \rightarrow \mathcal{F}''$  be the epimorphism from the extension. By injectivity of  $\mathcal{I}^0$ , there exists a morphism  $\beta : \mathcal{F} \rightarrow \mathcal{I}^0$  that extends  $\mathcal{F}' \rightarrow \mathcal{I}^0$ , which induces a morphism  $\gamma : \mathcal{F}'' \rightarrow \mathcal{I}^1$  making the following diagram commute

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}' & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{F}'' \longrightarrow 0 \\ & & \downarrow & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & \mathcal{F}' & \longrightarrow & \mathcal{I}^0 & \longrightarrow & \mathcal{J} \longrightarrow 0 \end{array}.$$

The pull-back  $\mathcal{I}^0 \times_{\mathcal{I}^1} \mathcal{F}''$  is unique up to unique isomorphism, so it remains to show  $\gamma$  is a representative of  $\xi$ . Again,  $\text{Hom}(\mathcal{F}'', \cdot)$  is a covariant  $\delta$ -functor with  $\text{Hom}(\mathcal{F}'', \mathcal{F}'') \rightarrow \text{Hom}(\mathcal{F}'', \mathcal{J})$  given by composition with  $\gamma$ , so  $\gamma$  is a representative of  $\xi$ .

□

3. Let  $X$  be a Noetherian scheme, and let  $\mathcal{F}, \mathcal{G} \in \mathfrak{Mod}(X)$ .

- (a) If  $\mathcal{F}, \mathcal{G}$  are both coherent, then  $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$  is coherent, for all  $i \geq 0$ .
- (b) If  $\mathcal{F}$  is coherent and  $\mathcal{G}$  is quasi-coherent, then  $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$  is quasi-coherent, for all  $i \geq 0$ .

*Proof.*

- (a) The question is local (6.2), so assume  $X = \text{Spec } A$  for some Noetherian ring  $A$ , and  $\mathcal{F} \cong \tilde{M}, \mathcal{G} \cong \tilde{N}$  for some  $A$ -modules  $M, N$ . Because  $A$  is Noetherian, there exists a finite free resolution of  $M$

$$F_\bullet \longrightarrow M \longrightarrow 0$$

where each  $F_\bullet$  is a free  $A$ -module of finite rank. Applying the  $\sim$ -functor and setting  $\mathcal{L}_\bullet = \tilde{F}_\bullet$ , we obtain a free resolution of  $\mathcal{F}$  of finite rank

$$\mathcal{L}_\bullet \longrightarrow \mathcal{F} \longrightarrow 0,$$

which we can use to compute  $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$  (6.5)

$$\mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \cong h^i(\mathcal{H}om(\mathcal{L}_\bullet, \mathcal{G})).$$

By equivalence of categories between coherent  $\mathcal{O}_X$ -modules and finitely generated  $A$ -modules (§2, 5.5), one has

$$\mathcal{H}om(\mathcal{L}_\bullet, \mathcal{G}) \cong \widehat{\text{Hom}_A(F_\bullet, N)}.$$

Each  $\mathcal{H}om(\mathcal{L}_\bullet, \mathcal{G})$  is coherent because each  $\text{Hom}(F_\bullet, N)$  is finitely generated, and  $A$  is Noetherian, so any sub- $\mathcal{O}_X$ -module of  $\mathcal{H}om(\mathcal{L}_\bullet, \mathcal{G})$  is also coherent (§2, 5.7). Hence,  $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$  is coherent for every  $i \geq 0$ .

- (b) The proof is essentially the same as (a) except  $\mathcal{G} \cong \tilde{N}$  for some  $A$ -module  $N$ , and the result still follows by (§2, 5.7).

□

6. Let  $A$  be a regular local ring, and let  $M$  be a finitely generated  $A$ -module. In this case, strengthen the result (6.10A) as follows.

- (a)  $M$  is projective if and only if  $\text{Ext}^i(M, A) = 0$  for all  $i > 0$ .
- (b) Use (a) to show that for any  $n$ ,  $\text{pd } M \leq n$  if and only if  $\text{Ext}^i(M, A) = 0$  for all  $i > n$ .

*Proof.*

- (a) Noetherian local rings have finite Krull dimension, so  $n = \text{pd}(M) < \infty$  (6.11A), and  $\text{Ext}_A^i(M, N) = 0$  for all  $i > n$  and all  $A$ -modules  $N$  (6.10A). Suppose  $\text{Ext}_A^i(M, A) = 0$  for all  $i > 0$ . Let  $N$  be any finitely generated  $A$ -module. There exists an exact sequence

$$0 \longrightarrow K \longrightarrow A^{\oplus m} \longrightarrow N \longrightarrow 0,$$

where  $K$  is some finitely generated  $A$ -module. Applying  $\text{Hom}_A(M, \cdot)$  gives a long exact sequence in cohomology

$$\begin{aligned} 0 &\longrightarrow \text{Hom}_A(M, K) \longrightarrow \text{Hom}_A(M, A^{\oplus m}) \longrightarrow \text{Hom}_A(M, N) \longrightarrow \\ &\longrightarrow \text{Ext}_A^1(M, K) \longrightarrow \text{Ext}_A^1(M, A^{\oplus m}) \longrightarrow \text{Ext}_A^1(M, N) \longrightarrow \cdots \\ &\cdots \longrightarrow \text{Ext}_A^n(M, K) \longrightarrow \text{Ext}_A^n(M, A^{\oplus m}) \longrightarrow \text{Ext}_A^n(M, N) \longrightarrow 0. \end{aligned}$$

The  $\text{Hom}_A(M, \cdot)$  functor commute with direct sums, so  $\text{Ext}_A^i(M, A^{\oplus m}) \cong \text{Ext}_A^i(M, A)^{\oplus m} = 0$  for all  $i > 0$ . By the long exact sequence of above,  $\text{Ext}_A^{i-1}(M, N) \cong \text{Ext}_A^i(M, K)$  and  $\text{Ext}_A^n(M, N) = 0$ . Thus,  $\text{Ext}_A^n(M, N) = 0$  for all finitely generated  $A$ -modules  $N$ , and descending induction on  $i$  shows  $\text{Ext}_A^i(M, N) = 0$  for all  $i > 0$  and all finitely generated  $A$ -modules  $N$ .

By assumption,  $M$  is finitely generated, so there exists an exact sequence

$$0 \longrightarrow Q \longrightarrow A^{\oplus r} \longrightarrow M \longrightarrow 0$$

where  $Q$  is a finitely generated  $A$ -module. Since  $\text{Ext}^1(M, Q) = 0$ , which classifies all extensions of  $M$  by  $Q$ ,  $A^{\oplus r}$  must be the trivial extension of  $M$  by  $Q$ , i.e.,  $A^{\oplus r} \cong M \oplus Q$ .

Let  $\pi : N' \rightarrow N$  be a surjective  $A$ -module homomorphism and  $\phi : M \rightarrow N$  any  $A$ -module homomorphism. There is a natural extension of  $\phi$  to  $A^{\oplus r}$  via taking the pull-back by the projection  $A^{\oplus r} \rightarrow M$ . Any free module is projective, so there exists unique  $A$ -module homomorphism  $h : A^{\oplus r} \rightarrow N'$  such that  $\phi' = \pi \circ h$ , so by restriction one has  $\phi = \pi \circ h|_M$ . Hence,  $M$  is projective. In particular, a module is projective if and only if it is a direct summand of a free module.

(b) We prove the following general result from (Weibel, 1994):

**Lemma 0.1.** *Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a left exact contravariant functor between two Abelian categories. Assume  $\mathcal{A}$  has enough projectives. If*

$$0 \longrightarrow N \longrightarrow P_m \longrightarrow \cdots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

*is an exact sequence in  $\mathcal{C}$  with the  $P_i$  projective (or  $F$ -cyclic), then  $R^i F(M) \cong R^{i-m-1} F(N)$  for  $i \geq m+2$ .*

For each  $0 \leq j \leq m+1$ , label the morphisms  $f_j : P_j \rightarrow P_{j-1}$ , where  $P_{-1}, P_{m+1}$  understood as  $M, N$ , respectively. We have an exact sequence

$$0 \longrightarrow \text{im } f_{j+1} \longrightarrow P_j \longrightarrow \text{im } f_j \longrightarrow 0,$$

which induces the natural isomorphism

$$R^i F(\text{im } f_j) \cong R^{i-1} F(\text{im } f_{j+1})$$

for all  $i \geq 2$ . Thus,

$$R^i F(\text{im } f_j) \cong R^{i-k} F(\text{im } f_{j+k})$$

for all  $i \geq k+1$ . Since  $\text{im } f_0 = M$  and  $\text{im } f_{m+1} = N$ , setting  $j = 0$  and  $k = m+1$ , we obtain our desired result.

Continuing on to our main problem, let  $M$  be a finitely generated  $A$ -module, and suppose  $\text{Ext}^i(M, A) = 0$  for all  $i > n$ . Let  $P_\bullet \rightarrow M \rightarrow 0$  be any projective resolution of  $M$ , and consider the exact sequence

$$0 \longrightarrow N \longrightarrow P_{n-1} \longrightarrow \cdots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

where  $N$  is the image of  $P_{n+1} \rightarrow P_n$ . Applying Lemma 1 with  $F = \text{Hom}(\cdot, A)$  and  $i = n+1$  so that  $R^i F = \text{Ext}^i(\cdot, A)$ , one has an isomorphism

$$\text{Ext}^{n+1}(M, A) \cong \text{Ext}^1(N, A).$$

But  $\text{Ext}^{n+1}(M, A) = 0$ , so  $N$  is projective by (a). Hence,  $\text{pd } M \leq n$ .

□

7. Let  $X = \text{Spec } A$  be an affine Noetherian scheme. Let  $M, N$  be  $A$ -modules, with  $M$  finitely generated. Then

$$\text{Ext}_X^i(\tilde{M}, \tilde{N}) \cong \text{Ext}_A^i(M, N)$$

and

$$\mathcal{E}\mathcal{X}\mathcal{T}_X^i(\tilde{M}, \tilde{N}) \cong \widetilde{\text{Ext}_A^i(M, N)}.$$

*Proof.* Both sides are  $\delta$ -functors in  $N$  from  $\mathfrak{Mod}(A)$  to  $\mathfrak{Ab}$ , and the functors  $\text{Hom}_X(\tilde{M}, \cdot)$  and  $\text{Hom}_A(M, \cdot)$  are equal (§2, 5.5), so their derived functors (as functors from  $\mathfrak{Mod}(A)$  to  $\mathfrak{Ab}$ ) are the same.

For the next isomorphism, both sides are  $\delta$ -functors in  $N$  from  $\mathfrak{Mod}(A)$  to  $\mathfrak{Mod}(X)$  because the  $\sim$ -functor is an exact functor. Both sides vanish for  $i > 0$  and  $N$  injective  $A$ -module because localization is exact, and  $\tilde{N}$  is flasque

if  $N$  is injective (3.4). Thus, to invoke (1.3A), we need to show the functors are equal for  $i = 0$ . We check locally. If  $x \in X$  is a point, let  $\mathfrak{p}$  be the prime ideal in  $A$  corresponding to  $x$ . By (§2, 5.1) and (6.8), we have

$$\mathcal{E}xt_X^i(\tilde{M}, \tilde{N})_x \cong \text{Ext}_{A_{\mathfrak{p}}}^i(M_{\mathfrak{p}}, N_{\mathfrak{p}}), \quad \widetilde{\text{Ext}_A^i(M, N)}_x \cong \text{Ext}^i(M, N)_{\mathfrak{p}},$$

and there is a natural map of  $\mathcal{O}_X$ -modules

$$\begin{aligned} \widetilde{\text{Hom}_A(M, N)} &\rightarrow \mathcal{H}om_X(\tilde{M}, \tilde{N}) \\ f/s &\mapsto (m \mapsto f(m)/s). \end{aligned}$$

Thus, we reduce to the following algebraic problem: let  $A$  be a Noetherian ring,  $M, N$   $A$ -modules with  $M$  finitely generated. Then the natural map of  $A_{\mathfrak{p}}$ -modules

$$\text{Hom}_A(M, N)_{\mathfrak{p}} \rightarrow \text{Hom}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}, N_{\mathfrak{p}})$$

is an isomorphism. If  $M$  is finitely generated, there exists an exact sequence

$$A^{\oplus n} \longrightarrow A^{\oplus r} \longrightarrow M \longrightarrow 0.$$

Applying  $\text{Hom}_A(\cdot, N)$  and localizing the first exact sequence gives

$$0 \longrightarrow \text{Hom}_A(M, N)_{\mathfrak{p}} \longrightarrow \text{Hom}_A(A^{\oplus r}, N)_{\mathfrak{p}} \longrightarrow \text{Hom}_A(A^{\oplus n}, N)_{\mathfrak{p}}$$

Switching the order, i.e., localizing first then applying  $\text{Hom}_{A_{\mathfrak{p}}}(\cdot, N_{\mathfrak{p}})$ , gives

$$0 \longrightarrow \text{Hom}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}, N_{\mathfrak{p}}) \longrightarrow \text{Hom}_{A_{\mathfrak{p}}}(A_{\mathfrak{p}}^{\oplus r}, N_{\mathfrak{p}}) \longrightarrow \text{Hom}_{A_{\mathfrak{p}}}(A_{\mathfrak{p}}^{\oplus n}, N_{\mathfrak{p}}).$$

The statement obvious holds For finite free modules

$$\text{Hom}_A(A^{\oplus r}, N)_{\mathfrak{p}} \cong N_{\mathfrak{p}}^{\oplus r} \cong \text{Hom}_{A_{\mathfrak{p}}}(A_{\mathfrak{p}}^{\oplus r}, N_{\mathfrak{p}})$$

We obtain our desired isomorphism by applying the five lemma to the following commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}_A(M, N)_{\mathfrak{p}} & \longrightarrow & \text{Hom}_A(A^{\oplus r}, N)_{\mathfrak{p}} & \longrightarrow & \text{Hom}_A(A^{\oplus n}, N)_{\mathfrak{p}} \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \text{Hom}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}, N_{\mathfrak{p}}) & \longrightarrow & \text{Hom}_{A_{\mathfrak{p}}}(A_{\mathfrak{p}}^{\oplus r}, N_{\mathfrak{p}}) & \longrightarrow & \text{Hom}_{A_{\mathfrak{p}}}(A_{\mathfrak{p}}^{\oplus n}, N_{\mathfrak{p}}) \end{array}.$$

□

8. Prove the following theorem of Kleiman: if  $X$  is a Noetherian, integral, separated, locally factorial scheme, then every coherent sheaf on  $X$  is a quotient of a locally free sheaf (of finite rank).

- First show that open sets of the form  $X_s$ , for various  $s \in \Gamma(X, \mathcal{L})$ , and various invertible sheaves  $\mathcal{L}$  on  $X$ , form a base for the topology of  $X$ .
- Now use (II, 5.14) to show that any coherent sheaf is a quotient of a direct sum  $\bigoplus \mathcal{L}_i^{n_i}$  for various invertible sheaves  $\mathcal{L}_i$  and various integers  $n_i$ .

#### 10. Duality for a Finite Flat Morphism.

- Let  $f : X \rightarrow Y$  be a finite morphism of Noetherian schemes. For any quasi-coherent  $\mathcal{O}_Y$ -module  $\mathcal{G}$ ,  $\mathcal{H}om_Y(f_*\mathcal{O}_X, \mathcal{G})$  is a quasi-coherent  $f_*\mathcal{O}_X$ -module, hence corresponds to a quasi-coherent  $\mathcal{O}_X$ -module, which we call  $f^!\mathcal{G}$  (II, Ex. 5.17e).
- Show that for any coherent  $\mathcal{F}$  on  $X$  and any quasi-coherent  $\mathcal{G}$  on  $Y$ , there is a natural isomorphism

$$f_* \mathcal{H}om_X(\mathcal{F}, f^!\mathcal{G}) \xrightarrow{\sim} \mathcal{H}om_Y(f_*\mathcal{F}, \mathcal{G}).$$

- For each  $i \geq 0$ , there is a natural map

$$\varphi_i : \text{Ext}_X^i(\mathcal{F}, f^!\mathcal{G}) \rightarrow \text{Ext}_Y^i(f_*\mathcal{F}, \mathcal{G}).$$

